

## TAWAU, Malaysia

WMO: 964810

Lat: 4.313N

Lon: 118.122E

Elev: 17

StdP: 101.12

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 22.3       | 22.8 | 20.0                          | 14.7 | 26.5 | 20.8 | 15.5 | 25.5 | 5.6                   | 29.5 | 5.2 | 29.7 | 1.2                   | 330  | 0.301 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 7.2                    | 33.0            | 26.1 | 32.2 | 26.2 | 32.0 | 26.3 | 27.8                | 31.0 | 27.4 | 30.8 | 27.1 | 30.7 | 3.8                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.9                            | 22.7 | 30.6 | 26.2 | 21.7 | 29.9 | 26.1 | 21.5 | 29.8 | 89.3          | 30.8 | 87.0 | 31.5 | 85.8 | 30.9 | 32.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 5.5           | 5.0  | 4.4 | DB                         | 20.7 | 34.3               | 1.5 | 1.5                                                | 19.5 | 35.4       | 18.6 | 36.3       | 17.8 | 37.1       | 16.6 | 38.2 | (4) |
| (5)               |      |     | WB                         | 19.6 | 29.3               | 1.6 | 1.4                                                | 18.4 | 30.3       | 17.5 | 31.2       | 16.6 | 31.9       | 15.5 | 32.9 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.3   | 27.0  | 27.2  | 27.5  | 27.6  | 27.7  | 27.4  | 27.1  | 27.2  | 27.3  | 27.4  | 27.4  | 27.2 | (6)  |
| (7)  |                                                                                  | DBStd       | 0.80   | 0.84  | 0.88  | 0.89  | 0.71  | 0.75  | 0.80  | 0.72  | 0.74  | 0.76  | 0.74  | 0.65  | 0.81 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD10.0     | 6330   | 528   | 481   | 542   | 528   | 549   | 521   | 531   | 533   | 519   | 541   | 523   | 534  | (10) |
| (11) |                                                                                  | CDD18.3     | 3288   | 269   | 248   | 283   | 278   | 291   | 271   | 272   | 274   | 269   | 282   | 273   | 276  | (11) |
| (12) |                                                                                  | CDH23.3     | 31031  | 2323  | 2217  | 2690  | 2763  | 2952  | 2623  | 2554  | 2624  | 2597  | 2668  | 2527  | 2494 | (12) |
| (13) |                                                                                  | CDH26.7     | 10591  | 727   | 742   | 957   | 976   | 1049  | 898   | 844   | 889   | 910   | 934   | 856   | 810  | (13) |
| (14) | Wind                                                                             | WSAvg       | 1.7    | 1.6   | 1.8   | 1.8   | 1.7   | 1.6   | 1.6   | 1.7   | 1.9   | 1.7   | 1.6   | 1.5   | 1.5  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 2236   | 226   | 160   | 180   | 160   | 198   | 184   | 175   | 169   | 155   | 205   | 202   | 222  | (15) |
| (16) |                                                                                  | PrecMax     | 3005   | 339   | 348   | 373   | 322   | 291   | 361   | 324   | 254   | 271   | 368   | 282   | 346  | (16) |
| (17) |                                                                                  | PrecMin     | 1519   | 109   | 28    | 46    | 35    | 145   | 47    | 97    | 63    | 36    | 92    | 107   | 88   | (17) |
| (18) |                                                                                  | PrecStd     | 390    | 66    | 84    | 80    | 60    | 40    | 54    | 45    | 45    | 56    | 63    | 49    | 64   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 32.4  | 33.1  | 33.8  | 33.0  | 32.9  | 32.8  | 32.1  | 32.2  | 32.6  | 33.0  | 32.8  | 32.8 | (19) |
| (20) |                                                                                  |             | MCWB   | 25.9  | 25.6  | 25.0  | 26.3  | 27.0  | 26.6  | 26.3  | 26.2  | 26.3  | 26.3  | 26.6  | 26.5 | (20) |
| (21) |                                                                                  | 2%          | DB     | 31.7  | 32.1  | 32.7  | 32.1  | 32.1  | 32.0  | 31.4  | 31.8  | 32.0  | 32.1  | 32.0  | 31.9 | (21) |
| (22) |                                                                                  |             | MCWB   | 25.9  | 25.7  | 25.4  | 26.3  | 26.7  | 26.5  | 26.2  | 26.2  | 26.3  | 26.4  | 26.5  | 26.5 | (22) |
| (23) |                                                                                  | 5%          | DB     | 31.0  | 31.2  | 31.8  | 31.5  | 31.7  | 31.2  | 31.0  | 31.1  | 31.2  | 31.5  | 31.2  | 31.1 | (23) |
| (24) |                                                                                  |             | MCWB   | 25.8  | 25.7  | 25.6  | 26.2  | 26.6  | 26.4  | 26.1  | 26.2  | 26.2  | 26.4  | 26.4  | 26.3 | (24) |
| (25) |                                                                                  | 10%         | DB     | 30.2  | 30.6  | 31.0  | 31.0  | 31.0  | 30.8  | 30.3  | 30.5  | 30.8  | 30.9  | 30.8  | 30.5 | (25) |
| (26) |                                                                                  |             | MCWB   | 25.6  | 25.6  | 25.7  | 26.2  | 26.5  | 26.3  | 25.9  | 26.0  | 26.1  | 26.3  | 26.3  | 26.2 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 27.5  | 28.3  | 27.4  | 27.7  | 28.2  | 27.8  | 27.6  | 27.4  | 27.5  | 27.9  | 28.0  | 27.8 | (27) |
| (28) |                                                                                  |             | MCD B  | 30.8  | 30.6  | 31.1  | 31.0  | 31.2  | 31.2  | 30.5  | 30.7  | 31.0  | 31.4  | 31.2  | 31.3 | (28) |
| (29) |                                                                                  | 2%          | WB     | 26.8  | 26.7  | 26.7  | 27.2  | 27.5  | 27.2  | 26.9  | 26.7  | 26.9  | 27.1  | 27.2  | 27.1 | (29) |
| (30) |                                                                                  |             | MCD B  | 30.2  | 30.4  | 30.6  | 30.7  | 30.8  | 30.7  | 30.3  | 30.4  | 30.8  | 30.9  | 30.7  | 30.6 | (30) |
| (31) |                                                                                  | 5%          | WB     | 26.3  | 26.2  | 26.4  | 26.7  | 27.1  | 26.8  | 26.4  | 26.4  | 26.6  | 26.7  | 26.8  | 26.7 | (31) |
| (32) |                                                                                  |             | MCD B  | 29.7  | 29.9  | 30.1  | 30.3  | 30.6  | 30.2  | 29.9  | 30.0  | 30.3  | 30.4  | 30.3  | 30.1 | (32) |
| (33) |                                                                                  | 10%         | WB     | 25.9  | 25.9  | 26.0  | 26.4  | 26.7  | 26.4  | 26.1  | 26.1  | 26.2  | 26.4  | 26.4  | 26.3 | (33) |
| (34) |                                                                                  |             | MCD B  | 29.3  | 29.4  | 29.7  | 29.9  | 30.1  | 29.8  | 29.5  | 29.6  | 29.8  | 29.9  | 29.8  | 29.5 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.1   | 7.4   | 7.6   | 7.3   | 7.2   | 7.1   | 7.0   | 7.2   | 7.4   | 7.5   | 7.4   | 7.1  | (35) |
| (36) |                                                                                  |             | MCD BR | 8.0   | 8.5   | 8.8   | 8.0   | 7.9   | 7.9   | 7.7   | 7.8   | 8.0   | 8.4   | 8.1   | 7.9  | (36) |
| (37) |                                                                                  |             | MCWBR  | 3.7   | 3.8   | 3.7   | 3.6   | 3.7   | 3.7   | 3.7   | 3.7   | 3.8   | 4.0   | 3.9   | 3.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 7.4   | 7.7   | 7.8   | 7.5   | 7.4   | 7.4   | 7.3   | 7.5   | 7.8   | 7.8   | 7.7   | 7.5  | (38) |
| (39) |                                                                                  |             | MCWBR  | 3.9   | 3.9   | 3.8   | 3.8   | 3.9   | 3.8   | 3.8   | 3.9   | 3.9   | 4.1   | 4.0   | 3.9  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.399  | 0.405 | 0.406 | 0.411 | 0.402 | 0.404 | 0.409 | 0.451 | 0.466 | 0.484 | 0.409 | 0.396 | (40) |      |
| (41) |                                                                                  | taud        | 2.529  | 2.491 | 2.484 | 2.464 | 2.512 | 2.514 | 2.497 | 2.354 | 2.303 | 2.249 | 2.485 | 2.538 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 924    | 923   | 917   | 896   | 884   | 870   | 870   | 849   | 851   | 840   | 907   | 920   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 107    | 114   | 115   | 115   | 106   | 104   | 107   | 126   | 135   | 143   | 111   | 105   | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 4.63   | 5.17  | 5.37  | 5.52  | 5.25  | 4.88  | 4.99  | 5.21  | 5.21  | 5.01  | 4.87  | 4.71  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.39   | 0.52  | 0.51  | 0.42  | 0.31  | 0.22  | 0.31  | 0.25  | 0.32  | 0.31  | 0.32  | 0.35  | (45) |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) Station Only           |  | +0.17 | N/A     | N/A    | +0.36   | +0.47 | +0.42 | N/A         | N/A     | +121    | +121    | (46) |  |
| (47) Regional (2 neighbors) |  | +0.16 | N/A     | N/A    | +0.33   | +0.40 | +0.30 | N/A         | N/A     | +94     | +94     | (47) |  |

Nomenclature: See separate page